FloodSignal

AI-Powered Flood Reporting and Monitoring Platform

Project idea
FloodSignal helps people report flood-related emergencies through text or voice. It uses speech-to-text, AI-based report analysis, GPS and map-based location selection to turn reports into structured information that can be reviewed through a dashboard.

1. Problem Statement

During floods, affected people may find it difficult to communicate what is happening, describe the severity of the situation, or share the exact incident location. Reports may arrive as unstructured messages, making them harder to review and prioritize.

2. Proposed Solution

FloodSignal provides a simple reporting interface where users can type a complaint or speak about an incident. The platform converts voice into text, analyses the report to extract useful details, associates it with a location, and displays submitted reports on a monitoring dashboard.

3. How It Works

Step	Process	What happens
1	**Report**	A user submits a flood complaint by text or voice.
2	**Speech-to-text**	Gnani AI Prisma converts spoken input into text, supporting multilingual reporting.
3	**AI analysis**	Groq analyses the report and extracts incident type, severity, road accessibility, summary and attention indicators.
4	**Location**	GPS can identify the user's current position. Users can select a different incident location using a landmark or map pin.
5	**Review & dashboard**	Users review report details before submission. Reports are stored and shown on a map for monitoring.

4. Key Features

- **Text and voice reporting:** Users can report incidents in a convenient way.
- **Multilingual speech recognition:** Gnani AI Prisma converts supported spoken languages into text.
- **AI-powered report analysis:** Groq extracts structured details, including incident type, severity and road accessibility.
- **GPS and map-based location:** Users can use GPS or identify the incident location with a landmark or map pin.
- **Duplicate report detection:** The system checks for similar reports to reduce repeated entries.
- **Monitoring dashboard:** Submitted reports and their locations can be reviewed together.

5. Technology Stack

Frontend	React, TypeScript, Tailwind CSS
Backend & database	Supabase and Supabase Edge Functions
Speech recognition	Gnani AI Prisma Speech-to-Text
AI analysis	Groq (openai/gpt-oss-20b)
Maps	Leaflet and OpenStreetMap
Deployment	Vercel

6. Expected Impact

FloodSignal aims to make flood reporting more accessible and organized. By combining voice input, AI-assisted structuring and location information, it can help response teams review incoming reports and identify cases that may need attention. AI-generated classifications are preliminary and should be reviewed by responsible personnel; the platform does not itself verify emergencies or dispatch rescue teams.

Project Demo: https://floodsignal0035.vercel.app